第 1 章　测试：e^+e^- 湮灭

1.1　e^+e^- 湮灭为强子：R 比值

$e^+e^- \to$ 强子的总截面是 QCD 中唯一一个“因子化”退化为平凡形式、而微扰论陈述可以在每一阶被严格证明的硬过程：末态完全求和，初态无色，整个强相互作用信息都包含在电磁流的两点关联函数之中。本节依次给出：约定与 $R = 12\pi \operatorname{Im}\Pi$（第 1.1.1 小节）；领头阶（第 1.1.2 小节）；维数正规化下的 $\mathcal{O}(\alpha_s)$ 实、虚修正及极点相消（第 1.1.3 小节）；欧氏区域两圈关联函数的完整计算（第 1.1.4 小节）；R 在微扰论各阶红外有限的定理及其证明（第 1.1.5 小节）；重整化群与解析延拓的高阶结构（第 1.1.6 小节）；KLN 定理在 $\mathcal{O}(\alpha_s)$ 的体现（第 1.1.7 小节）；Sterman–Weinberg 锥喷注截面（第 1.1.8 小节）。

1.1.1　约定：流关联函数、谱函数与 R

电磁流 $j^\mu = \sum_q e_q \bar{q}\gamma^\mu q$（以 e 为单位）。定义

$$\Pi^{\mu\nu}(q) = \mathrm{i}\int \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}q\cdot x}\langle 0|T\, j^\mu(x) j^\nu(0)|0\rangle = (q^\mu q^\nu - q^2 g^{\mu\nu})\,\Pi(q^2), \tag{1.1}$$

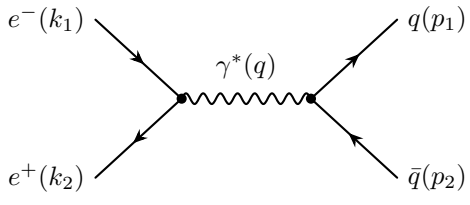
$\Pi^{\mu\nu}$ 的横向性来自流守恒。符号的取法使 $\operatorname{Im}\Pi > 0$；与 Peskin–Schroeder 的光子真空极化 Π_{PS}（$\Pi^{\mu\nu}_{\mathrm{PS}} = (q^2 g^{\mu\nu} - q^\mu q^\nu)\Pi_{\mathrm{PS}}$）的关系为：在 e 的最低阶，$\Pi_{\mathrm{PS}} = -e^2\Pi$。Wightman 函数（谱函数）为

$$\begin{aligned} W^{\mu\nu}(q) &= \int \mathrm{d}^4x\, \mathrm{e}^{\mathrm{i}q\cdot x}\langle 0|j^\mu(x) j^\nu(0)|0\rangle = \sum_X (2\pi)^4 \delta^4(q - p_X)\langle 0|j^\mu(0)|X\rangle\langle X|j^\nu(0)|0\rangle \\ &= 2\operatorname{Im}\Pi^{\mu\nu}(q) \qquad (q^0 > 0), \end{aligned} \tag{1.2}$$

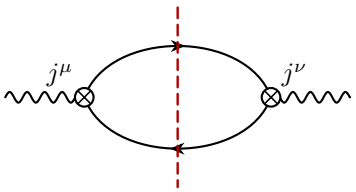
最后一式即 Källén–Lehmann 表示中的关系 $\operatorname{Im}\Pi = \pi\rho$；在下文中，它在 D 维中逐阶得到了验证（式 (1.6) 与 式 (1.19)）。对单光子交换、无质量轻子，自旋平均轻子张量满足 $L_{\mu\nu}(q^\mu q^\nu - q^2 g^{\mu\nu}) = s^2$，于是

$$\sigma(e^+e^- \to \text{强子}) = \frac{16\pi^2\alpha^2}{s}\operatorname{Im}\Pi(s + \mathrm{i}0), \qquad R(s) \equiv \frac{\sigma(e^+e^- \to \text{强子})}{\sigma(e^+e^- \to \mu^+\mu^-)} = 12\pi \operatorname{Im}\Pi(s + \mathrm{i}0), \tag{1.3}$$

其中 $\sigma_{\mu\mu} = 4\pi\alpha^2/(3s)$。在 $D = 4 - 2\varepsilon$ 维中只比较强子部分的量 $\rho(s) \equiv -g_{\mu\nu}W^{\mu\nu} = 2(D-1)s\operatorname{Im}\Pi(s)$；由于极点相消，$R^{(1)}/R_0 = \lim_{\varepsilon\to 0}(\rho_V + \rho_R)/\rho_0$ 与轻子部分如何延拓到 D 维无关。维数正规化约定：$\operatorname{Tr}\mathbb{1} = 4$，$g^\mu{}_\mu = D$，$c_\Gamma \equiv \Gamma(1+\varepsilon)\Gamma^2(1-\varepsilon)/\Gamma(1-2\varepsilon)$；裸耦合 $g_0^2 = 4\pi\alpha_s\mu_0^{2\varepsilon}$，$\overline{\mathrm{MS}}$ 中 $\mu_0^{2\varepsilon} \to \mu^{2\varepsilon}\mathrm{e}^{\varepsilon\gamma_E}(4\pi)^{-\varepsilon}$。



(a) 领头阶振幅



(b) $\Pi^{\mu\nu}$ 的单圈图及其割（式 (1.2)）

图 1.1　领头阶：$e^+e^- \to q\bar{q}$ 与电磁流关联函数

1.1.2　领头阶

对 $e^-(k_1)e^+(k_2) \to q(p_1)\bar{q}(p_2)$，$\sum|M|^2 = \frac{e^4 e_q^2 N_c}{s^2}\,\mathrm{Tr}[\not k_2\gamma^\mu \not k_1\gamma^\nu]\,\mathrm{Tr}[\not p_1\gamma_\mu \not p_2\gamma_\nu] = 8e^4e_q^2N_c\,(t^2+u^2)/s^2$，故 $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\alpha^2}{4s}N_c e_q^2(1+\cos^2\theta)$，

$$\sigma_0 = \frac{4\pi\alpha^2}{3s}N_c\sum_q e_q^2, \qquad R_0 = N_c\sum_q e_q^2 \quad \Big(=2,\ \tfrac{10}{3},\ \tfrac{11}{3}\ \text{对}\ n_f=3,4,5\Big). \tag{1.4}$$

D 维无质量两体相空间与单圈关联函数分别为

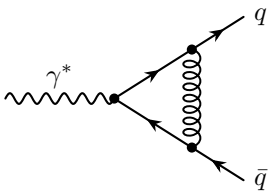
$$\Phi_2 = \frac{1}{8\pi}\Big(\frac{4\pi}{s}\Big)^{\varepsilon}\frac{\Gamma(1-\varepsilon)}{\Gamma(2-2\varepsilon)}, \qquad \Pi_0(q^2) = N_c e_q^2\,\frac{2(D-2)}{D-1}\,\frac{G(1,1)}{(4\pi)^{D/2}}\,(-q^2-\mathrm{i}0)^{-\varepsilon}, \tag{1.5}$$

$G(1,1) = \Gamma(\varepsilon)\Gamma^2(1-\varepsilon)/\Gamma(2-2\varepsilon)$。由 $-g_{\mu\nu}\,\mathrm{Tr}[\not p_1\gamma^\mu\not p_2\gamma^\nu] = 2(D-2)s$ 得 $\rho_0 = N_c e_q^2\,2(D-2)s\,\Phi_2$；利用 $(-s-\mathrm{i}0)^{-\varepsilon} = s^{-\varepsilon}\mathrm{e}^{\mathrm{i}\pi\varepsilon}$ 与 $\Gamma(\varepsilon)\Gamma(1-\varepsilon)\sin\pi\varepsilon = \pi$，

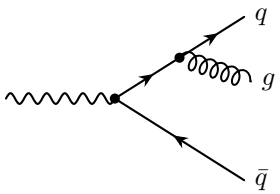
$$\rho_0 = 2(D-1)\,s\,\mathrm{Im}\,\Pi_0(s+\mathrm{i}0) \qquad \text{对一切}\varepsilon\ \text{精确成立}, \tag{1.6}$$

$D=4$ 时 $\mathrm{Im}\,\Pi_0 = N_c e_q^2/(12\pi)$，即 式 (1.4)。$\overline{\mathrm{MS}}$ 重整化后（第 1.1.4 小节）$\Pi_0 = \frac{N_c e_q^2}{12\pi^2}\big(\frac{5}{3}-L\big)$，$L \equiv \ln(-q^2/\mu^2)$。

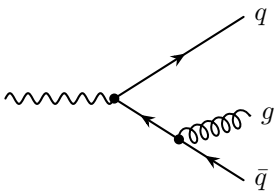
1.1.3　$\mathcal{O}(\alpha_s)$：维数正规化下的虚修正与实辐射



(a) 顶点修正



(b) 夸克辐射



(c) 反夸克辐射

图 1.2　$\mathcal{O}(\alpha_s)$ 的虚修正与实辐射（Feynman 规范；外线自能在维数正规化中为无标度积分，等于零）

一圈类时形状因子。　$\bar{u}(p_1)\Gamma^\mu v(p_2) = F(q^2)\,\bar{u}(p_1)\gamma^\mu v(p_2)$。对无质量在壳旋量，手征守恒只允许 γ^μ 结构，故 $F^{(1)} = \mathrm{Tr}[\not p_1\Lambda^\mu\not p_2\gamma_\mu]/\,\mathrm{Tr}[\not p_1\gamma^\mu\not p_2\gamma_\mu]$ 是精确投影。把标量积化为传播子分母后只

剩三个积分：三角形 $J(1,1,1)$、s 道泡 $J(0,1,1)$ 及 $J(-1,1,1)=-\frac{s}{2}J(0,1,1)$。结果对 ε 精确：

$$\begin{aligned} F^{(1)}(q^2) &= -\frac{\alpha_s C_F}{4\pi}\Big(\frac{4\pi\mu_0^2}{-q^2-\mathrm{i}0}\Big)^{\varepsilon} c_\Gamma\,\frac{2-\varepsilon+2\varepsilon^2}{\varepsilon^2(1-2\varepsilon)} \\ &= \frac{\alpha_s C_F}{4\pi}\Big(\frac{4\pi\mu_0^2}{-q^2-\mathrm{i}0}\Big)^{\varepsilon} c_\Gamma\Big[-\frac{2}{\varepsilon^2}-\frac{3}{\varepsilon}-8-16\varepsilon+\mathcal{O}\big(\varepsilon^2\big)\Big]. \end{aligned} \tag{1.7}$$

用 $\mathrm{e}^{\varepsilon\gamma_E}c_\Gamma = 1-\frac{\zeta_2}{2}\varepsilon^2-\frac{7}{3}\zeta_3\varepsilon^3+\cdots$ 化为 $\overline{\mathrm{MS}}$ 形式：

$$F^{(1)} = \frac{\alpha_s C_F}{4\pi}\Big[-\frac{2}{\varepsilon^2}+\frac{2L-3}{\varepsilon}-L^2+3L-8+\zeta_2+\mathcal{O}(\varepsilon)\Big], \qquad L=\ln\frac{-q^2-\mathrm{i}0}{\mu^2}, \tag{1.8}$$

$L=0$ 处的 $\mathcal{O}(\varepsilon)$ 项为 $-16+\frac{3}{2}\zeta_2+\frac{14}{3}\zeta_3$，与文献 [216, 217] 一致。类时区 $q^2=s+\mathrm{i}0$ 时 $L=\ell-\mathrm{i}\pi$，$\ell\equiv\ln(s/\mu^2)$：

$$\operatorname{Im}F^{(1)} = \frac{\alpha_s C_F}{4\pi}\Big[-\frac{2\pi}{\varepsilon}+2\pi\ell-3\pi\Big], \qquad 2\operatorname{Re}F^{(1)} = \frac{\alpha_s C_F}{2\pi}\Big[-\frac{2}{\varepsilon^2}+\frac{2\ell-3}{\varepsilon}-\ell^2+3\ell-8+\zeta_2+\pi^2\Big]. \tag{1.9}$$

发散的相位在 $|F|^2$ 中消去；由于 $\operatorname{Re}L^2=\ell^2-\pi^2$，类时值比同一 $|q^2|$ 的类空值多出 π^2。在维数正规化中外线自能为零，顶点图的紫外极点被（为零的）在壳 LSZ 因子的紫外部分抵消（Ward 恒等式 $Z_1=Z_2$），守恒流本身无需重整化：F 的全部极点都是红外的，$1/\varepsilon^2$ 来自软—共线区，$1/\varepsilon$ 来自软或共线区。

实辐射。　$M^\nu = -g_0 t^a \bar u(p_1)\big[\gamma^\alpha(\not p_1+\not k)\gamma^\nu/s_{13}-\gamma^\nu(\not p_2+\not k)\gamma^\alpha/s_{23}\big]v(p_2)\varepsilon^*_\alpha$；只有一个胶子时 $k_\alpha M^\alpha=0$，极化求和取 $-g_{\alpha\beta}$ 是精确的。令 $y_{ij}=2p_i\cdot p_j/s$（$y_{12}+y_{13}+y_{23}=1$），则

$$-g_{\mu\nu}\sum M^\mu M^{\nu *} = 8(1-\varepsilon)g_0^2 C_F N_c e_q^2\Big[(1-\varepsilon)\Big(\frac{y_{13}}{y_{23}}+\frac{y_{23}}{y_{13}}\Big)+\frac{2y_{12}}{y_{13}y_{23}}-2\varepsilon\Big], \tag{1.10}$$

D 维三体相空间为

$$\mathrm{d}\Phi_3 = \frac{(4\pi)^{2\varepsilon-3}s^{1-2\varepsilon}}{2\Gamma(2-2\varepsilon)}\,\mathrm{d}y_{13}\,\mathrm{d}y_{23}\,(y_{12}y_{13}y_{23})^{-\varepsilon}. \tag{1.11}$$

用 Dirichlet 积分 $\int[\mathrm{d}y]\,y_{12}^{a-1}y_{13}^{b-1}y_{23}^{c-1}=\Gamma(a)\Gamma(b)\Gamma(c)/\Gamma(a+b+c)$ 得到对 ε 精确的结果

$$\begin{aligned} \frac{\rho_R}{\rho_0} &= \frac{\alpha_s C_F}{2\pi}\Big(\frac{4\pi\mu_0^2}{s}\Big)^{\varepsilon}\frac{2(1-2\varepsilon)(2-2\varepsilon+\varepsilon^2)\Gamma^2(1-\varepsilon)}{\varepsilon^2(1-3\varepsilon)(2-3\varepsilon)\Gamma(1-3\varepsilon)} \\ &= \frac{\alpha_s C_F}{2\pi}\Big(\frac{4\pi\mu_0^2}{s}\Big)^{\varepsilon} c_\Gamma\Big[\frac{2}{\varepsilon^2}+\frac{3}{\varepsilon}+\frac{19}{2}-\pi^2+\mathcal{O}(\varepsilon)\Big]; \end{aligned} \tag{1.12}$$

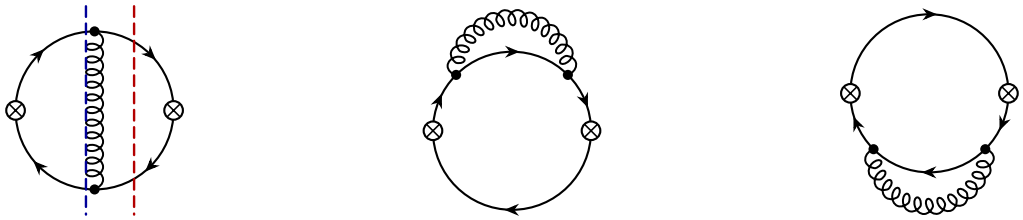
在四维中它就是熟知的 $\frac{1}{\sigma_0}\frac{\mathrm{d}\sigma}{\mathrm{d}x_1\mathrm{d}x_2}=\frac{C_F\alpha_s}{2\pi}\frac{x_1^2+x_2^2}{(1-x_1)(1-x_2)}$，$x_i=2E_i/\sqrt{s}$。

命题 1.1（$\mathcal{O}(\alpha_s)$ 的 R）　R严格 虚修正 $\rho_V/\rho_0=2\operatorname{Re}F^{(1)}=\frac{\alpha_s C_F}{2\pi}(\frac{4\pi\mu_0^2}{s})^{\varepsilon}c_\Gamma\cos(\pi\varepsilon)\,f(\varepsilon)$，$f=-(2-\varepsilon+2\varepsilon^2)/(\varepsilon^2(1-2\varepsilon))$，即 $\big[-\frac{2}{\varepsilon^2}-\frac{3}{\varepsilon}-8+\pi^2+\mathcal{O}(\varepsilon)\big]$。与 式 (1.12) 相加，

$$\frac{\rho_R+\rho_V}{\rho_0} = \frac{\alpha_s C_F}{2\pi}\Big(\frac{4\pi\mu_0^2}{s}\Big)^{\varepsilon}\Big[\frac{3}{2}+\mathcal{O}(\varepsilon)\Big] \quad\Longrightarrow\quad R = R_0\Big(1+\frac{3C_F}{4}\frac{\alpha_s}{\pi}\Big) = R_0\Big(1+\frac{\alpha_s}{\pi}\Big), \tag{1.13}$$

其中 $1/\varepsilon^2$ 与 $1/\varepsilon$ 极点精确相消 [86, 87]。

有限部分 $\frac{3}{2}=\frac{19}{2}-8$ 要求实、虚部分用同一归一化：两种 Γ 前因子在 $\mathcal{O}(\varepsilon^2)$ 不同，$\Gamma^2(1-\varepsilon)/\Gamma(1-3\varepsilon)=c_\Gamma(1-\frac{\pi^2}{2}\varepsilon^2+\cdots)$，正是这一差别把 $\frac{19}{2}$ 变成了 式 (1.12) 中的 $\frac{19}{2}-\pi^2$。式 (1.12) 与虚修正中方括号的形式与 Ellis–Stirling–Webber[4] 第 3 章相同。



(a) V（两种割）　(b) S_1　(c) S_2

图 1.3　$\Pi^{\mu\nu}$ 的 $\mathcal{O}(g^2)$ 图。V 中的红色割（两粒子）给出 $2\,\mathrm{Re}\,F^{(1)}$，蓝色割（三粒子）给出实辐射的干涉项；$S_{1,2}$ 的两粒子割经过在壳自能，在维数正规化中为零，其三粒子割给出实辐射的“对角”项

1.1.4　欧氏区域的两圈关联函数

三个图（图 1.3）都带因子 $\mathrm{i}\cdot(-1)\cdot(\mathrm{i}g)^2\cdot(-\mathrm{i})\cdot\mathrm{i}^4\cdot C_F N_c = g^2 C_F N_c$。令 $r=k+l$，分母属于标准两圈传播子拓扑

$$\begin{aligned} &D_1=k^2,\quad D_2=(k-q)^2,\quad D_3=r^2,\quad D_4=(r-q)^2,\quad D_5=(k-r)^2;\\ &\mathrm{V}:\ (a_i)=(1,1,1,1,1),\qquad \mathrm{S}_1:\ (2,1,1,0,1),\qquad \mathrm{S}_2:\ (1,2,0,1,1). \end{aligned} \tag{1.14}$$

Wick 转动后需要欧氏积分 $\hat I(a)=\int\!\!\int \frac{\mathrm{d}^D k\,\mathrm{d}^D r}{\pi^D}\prod_i D_i^{-a_i}$（$a_i$ 可正可负，系数为 $(Q^2)^{D-\sum a_i}$，$Q^2=-q^2$）。它们由三条规则精确算出：(i) G 函数 $\int \frac{\mathrm{d}^D k}{\pi^{D/2}}\frac{1}{(k^2)^a((k-q)^2)^b}=G(a,b)(Q^2)^{\frac{D}{2}-a-b}$，$G(a,b)=\frac{\Gamma(a+b-\frac{D}{2})\Gamma(\frac{D}{2}-a)\Gamma(\frac{D}{2}-b)}{\Gamma(a)\Gamma(b)\Gamma(D-a-b)}$；(ii) 单圈张量公式（Feynman 参数 $k=l+xp$）$\int \frac{\mathrm{d}^D k}{\pi^{D/2}}\frac{k^{\mu_1}\dots k^{\mu_m}}{(k^2)^\alpha((p-k)^2)^\beta}=\sum_t A_t(p^2)^{\frac{D}{2}+t-\alpha-\beta}\{g^t p^{m-2t}\}$，$A_t=\frac{\Gamma(\alpha+\beta-\frac{D}{2}-t)\Gamma(\frac{D}{2}-\alpha+m-t)\Gamma(\frac{D}{2}-\beta+t)}{2^t\Gamma(\alpha)\Gamma(\beta)\Gamma(D-\alpha-\beta+m)}$；(iii) 分部积分的三角形规则 [224, 225]

$$(D-a_1-a_2-2a_5)\,\hat I = a_1\big[\hat I(1^+5^-)-\hat I(1^+3^-)\big]+a_2\big[\hat I(2^+5^-)-\hat I(2^+4^-)\big], \tag{1.15}$$

例如 $\hat I(1,1,1,1,1)=-\frac{1}{\varepsilon}G(1,1)[G(2,1)-G(2,1+\varepsilon)]=6\zeta_3+\mathcal{O}(\varepsilon)$。计算表明 $q_\mu q_\nu\Pi^{\mu\nu}=0$ 在一圈与两圈都对 ε 精确成立（对迹、动量走向与积分的强检验），并且

$$\begin{aligned} \Pi^{(1)}_{\mathrm{bare}} &= N_c e_q^2\,C_F\,\frac{g_0^2(Q^2)^{-2\varepsilon}}{(4\pi)^D}\Big[r_1\,G(1,1)^2+r_2\,G(1,1)\,G(1{+}\varepsilon,1)\Big],\\ r_1=-\frac{4(1-\varepsilon)(2-\varepsilon+2\varepsilon^2)}{\varepsilon(3-2\varepsilon)},&\qquad r_2=\frac{16(1-\varepsilon)(1-2\varepsilon)(2-2\varepsilon+\varepsilon^2)}{\varepsilon(3-2\varepsilon)(2-3\varepsilon)}, \end{aligned} \tag{1.16}$$

$G(1{+}\varepsilon,1)=\Gamma(2\varepsilon)\Gamma(1-2\varepsilon)\Gamma(1-\varepsilon)/[\Gamma(1+\varepsilon)\Gamma(2-3\varepsilon)]$。$r_1$ 含有形状因子的多项式 $2-\varepsilon+2\varepsilon^2$，$r_2$ 含有实辐射的多项式 $2-2\varepsilon+\varepsilon^2$：两种超越结构分别对应两粒子割与三粒子割。以 $\frac{\alpha_s}{\pi}C_F\frac{N_c e_q^2}{12\pi^2}$ 为单位并乘以 $\overline{\mathrm{MS}}$ 因子，展开为

$$\mathrm{V}=\frac{1}{4\varepsilon^2}+\frac{\frac{17}{12}-\frac{L}{2}}{\varepsilon}+\cdots,\qquad \mathrm{S}_1{+}\mathrm{S}_2=-\frac{1}{4\varepsilon^2}+\frac{\frac{L}{2}-\frac{25}{24}}{\varepsilon}+\cdots,\qquad \text{和}=\frac{3}{8\varepsilon}+\frac{55}{16}-3\zeta_3-\frac{3}{4}L+\mathcal{O}(\varepsilon):$$

单个图中的 $1/\varepsilon^2$ 与非局域的 L/ε 在和中消去（子发散按 $Z_1=Z_2$ 相消），剩下的是*局域*单极点，由 Π 的加法（接触项）抵消项吸收。于是

$$\Pi_{\overline{\mathrm{MS}}}(q^2)=\frac{N_c e_q^2}{12\pi^2}\Big[\frac{5}{3}-L+\frac{\alpha_s}{\pi}\,\frac{3C_F}{4}\Big(\frac{55}{12}-4\zeta_3-L\Big)\Big]+\mathcal{O}\big(\alpha_s^2\big),\qquad L=\ln\frac{-q^2}{\mu^2}, \tag{1.17}$$

常数项依赖于方案（接触项），L 的系数不依赖；常数 $\frac{55}{12}-4\zeta_3$ 与文献 [204, 212] 一致（引用时须注明归一化：$\frac{3}{16\pi^2}(\frac{20}{9}-\frac{4}{3}L)$ 等于 $N_c=3$ 时的 $\frac{N_c}{12\pi^2}(\frac{5}{3}-L)$）。抵消项 $Z=z_0/\varepsilon+\alpha_s z_1/\varepsilon+\cdots$ 给出非齐次重整化群方程与 Adler 函数 [211]

$$\mu^2\frac{\mathrm{d}\Pi}{\mathrm{d}\mu^2}=z_0+2z_1\alpha_s=\frac{N_c e_q^2}{12\pi^2}\Big(1+\frac{3C_F}{4}\frac{\alpha_s}{\pi}\Big),\qquad D(Q^2)\equiv-12\pi^2Q^2\frac{\mathrm{d}\Pi}{\mathrm{d}Q^2}=N_c e_q^2\Big(1+\frac{3C_F}{4}\frac{\alpha_s}{\pi}\Big),\tag{1.18}$$

D 有限、与 μ 无关，完全在欧氏区域定义（取 $C_F\to1$ 即得两圈 QED β 函数中的 $1+\frac{3}{4}\frac{\alpha}{\pi}$）。$\Pi$ 在 $q^2\in\mathbb{C}\setminus[0,\infty)$ 解析，$q^2=s+\mathrm{i}0$ 处 $L\to\ell-\mathrm{i}\pi$，故 $R=12\pi\,\mathrm{Im}\,\Pi=R_0(1+\alpha_s/\pi)$，与 命题 1.1 相同。

命题 1.2（D 维中精确的割图恒等式）　R 严格　由 $(Q^2)^{-2\varepsilon}\to s^{-2\varepsilon}\mathrm{e}^{2\pi\mathrm{i}\varepsilon}$，

$$2(D-1)\,s\;\mathrm{Im}\,\Pi^{(1)}_{\mathrm{bare}}(s+\mathrm{i}0)=\rho_V+\rho_R\qquad\text{对一切}\varepsilon\text{ 成立.}\tag{1.19}$$

证明　两边均已化为 Γ 函数的闭式（式 (1.16) 与 式 (1.7)与(1.12)）。利用 $(-s-\mathrm{i}0)^{-2\varepsilon}=s^{-2\varepsilon}\mathrm{e}^{2\pi\mathrm{i}\varepsilon}$、$\sin2\pi\varepsilon=2\sin\pi\varepsilon\cos\pi\varepsilon$ 与反射公式 $\Gamma(z)\Gamma(1-z)=\pi/\sin\pi z$（$z=\varepsilon,2\varepsilon$），可以逐项证明：$r_1G(1,1)^2$ 一项恰好等于虚修正（两粒子割，含 $\cos\pi\varepsilon$），$r_2G(1,1)G(1{+}\varepsilon,1)$ 一项恰好等于实辐射（三粒子割，含 $\Gamma(1-3\varepsilon)$）；两项之比经符号化简都恒等于 1，此外还在若干个 ε 值上以 30 位精度作了数值核对。 □

这是 $\mathcal{O}(g^2)$ 的幺正性在 D 维中的直接验证：欧氏两圈积分与闵氏相空间积分这两个完全独立的计算在取 $\varepsilon\to0$ 之前就已一致。

1.1.5　R 在微扰论各阶的红外有限性

定理 1.3（R 的红外有限性）　R 严格（作为 $12\pi\,\mathrm{Im}\,\Pi$ 的陈述）　A 全阶论证（与 D 维部分子割图之和的等同）

在 n_f 味无质量 QCD 中（$\overline{\mathrm{MS}}$ 重整化），令 $R^{(n)}(s;\varepsilon)$ 为 a^n（$a=\alpha_s(\mu)/\pi$）的系数：它是 $\gamma^*\to$ 部分子的全部割图之和（等价地，对全部物理部分子末态求和），圈积分与相空间都取 D 维、包含 $\overline{\mathrm{MS}}$ 抵消项插入，并以 ρ_0 归一。则对每个 n：

(i) $\lim_{\varepsilon\to0}R^{(n)}(s;\varepsilon)$ 存在；

(ii) $n\ge1$ 时该极限是 $\ln(s/\mu^2)$ 的次数不超过 $n-1$ 的多项式（$R^{(0)}=R_0$）；

(iii) $R=12\pi\,\mathrm{Im}\,\Pi_{\overline{\mathrm{MS}}}(s+\mathrm{i}0)$，其中 $\Pi_{\overline{\mathrm{MS}}}$ 是欧氏区域的重整化关联函数的解析延拓。

证明用到以下三个结果，以下给出精确陈述与来源。

(A) 非例外欧氏动量处的红外有限性。在只含无质量场、且没有超可重整耦合的可重整理论中，若紫外重整化不在零动量处作减除（$\overline{\mathrm{MS}}$，或带辅助质量的 BPHZL），则在非例外欧氏动量（外动量的任何真部分和都不为零）处，重整化格林函数在每一阶有限。物理层面的一般证明见 Poggio–Quinn[75]（其推论“物理不连续性亦红外有限”正是此处所用）；数学上严格的形式见 Lowenstein–Zimmermann[58]（带无质量传播子的幂次计数定理）、Lowenstein[233]（BPHZL 收敛定理），以及维数重整化中的 Breitenlohner–Maison[231, 232]（无质量粒子理

论中维数重整化格林函数在各阶的存在性；后者的条件是相互作用的红外次数 $\Omega \geq 4$）；无质量 ϕ^4 理论红外结构的早期分析见 Symanzik[235]。应用于 Π：在 e 的最低阶、g 的所有阶，$\Pi^{\mu\nu}$ 就是无质量 QCD×QED 中光子的自能，即一个只含四维耦合的可重整无质量理论中基本场的格林函数；两点函数在 $q \neq 0$ 处自动是非例外的。（待核：Breitenlohner–Maison 的假设对协变规范下带 Faddeev–Popov 鬼场的非阿贝尔规范场是否已逐条写明。幂次计数是相同的：QCD 的每个顶点量纲为 4，胶子传播子连同 $(1-\xi)k_\mu k_\nu/k^4$ 部分的红外次数为 -2；规范不变流的关联函数与规范无关。）

(B) 无质量传播子型积分的结构。由量纲分析，每个裸的 $(n+1)$ 圈贡献都等于 $c_n(\varepsilon)(\mu_0^2)^{n\varepsilon}(-q^2-\mathrm{i}0)^{-(n+1)\varepsilon}$，$c_n$ 是 ε 的亚纯函数（例如 式 (1.5)与(1.16)）；$\overline{\mathrm{MS}}$ 抵消项是 $1/\varepsilon$ 的多项式 [226]（维数重整化的一般框架与作用量原理见 [230]），维数正规化保持规范不变性 [64]。因此在固定阶，$\Pi_{\overline{\mathrm{MS}}}(q^2;\varepsilon)$ 的每个 Laurent 系数都是 $L=\ln(-q^2/\mu^2)$ 的多项式。

(C) D 维中的幺正性（割图规则）。对每个图，把两个流顶点分开的全部割之和等于该图的不连续性（Cutkosky[59]；Veltman 的最大时间方程 [60]；'t Hooft–Veltman[228]）。协变规范下鬼场割与非物理极化割相互抵消（'t Hooft–Veltman[227]；BRST，Kugo–Ojima[229]），故割图之和等于对物理末态的求和。该恒等式对维数正规化积分作为 ε 的亚纯函数之间的恒等式成立，在 $\mathcal{O}(g^2)$ 由 命题 1.2 精确验证。Π 的加法抵消项是实的且与 q 无关，没有不连续性；耦合常数抵消项对应于割图中的抵消项顶点插入。

定理 1.3 的证明　(a) 由 (A)，对 $q^2<0$，重整化的 $\Pi^{(n)}_{\overline{\mathrm{MS}}}(q^2;\varepsilon)$ 在 $\varepsilon\to 0$ 时对一切实的 L 有有限极限。(b) 由 (B)，每个极点系数 $p_{n,k}(L)$（$k<0$）都是 L 的多项式，并且对一切实 L 为零，因而恒等于零，在复数 L 处也为零，特别是在 $L=\ell-\mathrm{i}\pi$ 处。于是 $\Pi^{(n)}_{\overline{\mathrm{MS}}}(s+\mathrm{i}0;\varepsilon)$（在固定 ε 下由裸表达式 $(-q^2-\mathrm{i}0)^{-(n+1)\varepsilon}$ 延拓定义）有有限极限，它等于把 $L=\ell-\mathrm{i}\pi$ 代入同一多项式所得，其虚部是 ℓ 的实系数多项式。正是在这一步，$\varepsilon\to 0$ 与解析延拓可以交换。(c) 由 (C)，在每个 ε 下 $\rho^{(n)}(s;\varepsilon)=2(D-1)s\,\mathrm{Im}\,\Pi^{(n)}_{\overline{\mathrm{MS}}}(s+\mathrm{i}0;\varepsilon)$，故 $R^{(n)}(s;\varepsilon)$ 有有限极限 $12\pi\,\mathrm{Im}\,\Pi^{(n)}_{\overline{\mathrm{MS}}}(s+\mathrm{i}0)$。次数的陈述 (ii) 由重整化群得到（第 1.1.6 小节）。 □

注 1.4（严格性的界定）　(a)(b) 两步在所引定理的意义下是 R 严格。(c) 把 $12\pi\,\mathrm{Im}\,\Pi$ 与 D 维部分子割图之和等同，标为 A 全阶论证：割图规则与 BRST 幺正性是标准的，但就作者所知，没有单一文献把 D 维、无质量、带红外正规化的形式写到完全严格。若直接以 $12\pi\,\mathrm{Im}\,\Pi$ 定义 R（由完整理论的幺正性，它就是物理截面），则“R 的微扰展开各阶红外有限”是 R 严格的陈述。闵氏空间中的独立全阶论证见 Sterman[221, 78, 79]（割真空极化图中质量发散的来源与相消）。Lowenstein–Speer[234] 证明了重整化 Feynman 积分（包括零质量情形）的 $\mathrm{i}0\to 0$ 极限作为 Lorentz 不变的缓增分布存在，故 $\mathrm{Im}\,\Pi$ 逐阶作为分布有定义；对无质量两点函数，显式形式使它在 $s>0$ 处是普通函数。若改用夸克质量作红外正规化，非例外欧氏点处没有质量奇异性（参见 [72, 236]），欧氏关联函数在固定 Q^2 下有光滑的 $m\to 0$ 极限，因而无质量的 R 也是有质量 R 的 $m\to 0$ 极限（A 全阶论证，此处未重新推导）。

注 1.5（微扰论之外）　$R(s)$ 本身含共振结构。与实验比较需要涂抹 [88]，或者改用欧氏量（Adler 函数）与算符乘积展开 [89]。对无质量夸克，主导幂次修正为 $\langle\frac{\alpha_s}{\pi}G^2\rangle/Q^4$（量纲 4；夸克凝聚只以 $m_q\langle\bar qq\rangle$ 的形式出现）。OPE 在非微扰意义下的成立以及局域对偶性属于 C 猜想。

1.1.6　重整化群与解析延拓：π^2 项

令 $a=\alpha_s/\pi$，$\mu^2\mathrm{d}a/\mathrm{d}\mu^2=-b_0a^2-b_1a^3-\cdots$，$b_0=\beta_0/4$，$b_1=\beta_1/16$；Adler 函数 $D=R_0\,d(a(Q^2))$，$d=1+d_1a+d_2a^2+\cdots$。由 $\mathrm{d}\Pi/\mathrm{d}\ln Q^2=-D/(12\pi^2)$，在 a^n 阶 $\Pi_{\overline{\mathrm{MS}}}$ 是 L 的 $\max(1,n)$ 次多项式。把 $\Pi(se^{-\mathrm{i}\pi})-\Pi(se^{\mathrm{i}\pi})$ 写成沿圆周的积分得到

$$R(s)=\frac{1}{2\pi}\int_{-\pi}^{\pi}\mathrm{d}\theta\,D(s\,\mathrm{e}^{\mathrm{i}\theta}),\tag{1.20}$$

它与直接代入 $L\to\ell-\mathrm{i}\pi$ 的结果逐项一致。以 $a(s)$ 展开 $R=R_0\sum r_na^n$，得

$$r_1=d_1,\quad r_2=d_2,\quad r_3=d_3-\frac{\pi^2}{3}b_0^2d_1,\quad r_4=d_4-\pi^2\Big(b_0^2d_2+\frac{5}{6}b_0b_1d_1\Big),\tag{1.21}$$

用 $a(\mu)$ 表示时，a^n 的系数是 $\ln(s/\mu^2)$ 的次数不超过 $n-1$ 的多项式（定理 1.3 (ii)）。对 SU(3)，$\frac{\pi^2}{3}b_0^2=16.66,\ 14.28,\ 12.09$（$n_f=3,4,5$）：这一纯运动学的“$\pi^2$ 项”构成已知 r_3 的主体 [213, 214, 215]。高阶系数见文献 [204, 205, 206, 207, 208, 209, 210]。在 a^3 出现正比于 $(\sum_q e_q)^2$ 的单态贡献：两个夸克圈之间的双胶子中间态被 C 不变性（Furry 定理，色因子 δ^{ab}）禁戒，三胶子给出 $d^{abc}d^{abc}$。

1.1.7　KLN 定理在 $\mathcal{O}(\alpha_s)$ 的体现

记 $N_\varepsilon=\frac{\alpha_sC_F}{2\pi}(\frac{4\pi\mu_0^2}{s})^\varepsilon c_\Gamma$。遍举截面各自发散：

$$\frac{\sigma(q\bar q)^{(1)}}{\sigma_0}=N_\varepsilon\Big[-\frac{2}{\varepsilon^2}-\frac{3}{\varepsilon}-8+\pi^2\Big],\qquad\frac{\sigma(q\bar qg)}{\sigma_0}=N_\varepsilon\Big[\frac{2}{\varepsilon^2}+\frac{3}{\varepsilon}+\frac{19}{2}-\pi^2\Big],\tag{1.22}$$

只有对简并末态（软胶子、共线分裂）求和才有限（式 (1.13)）。初态无色，因此只需对末态的简并态求和（Kinoshita[72]，Lee–Nauenberg[73]）；非阿贝尔 Bloch–Nordsieck 的反例 [103, 104] 涉及有色初态，与此处无关（一般情形参见 [223]）。

一个“单举”但不红外安全的量：夸克能谱。固定 $x=2E_q/\sqrt{s}<1$ 时，$q\parallel g$ 区被积分，留下共线极点

$$\frac{1}{\sigma_0}\frac{\mathrm{d}\sigma}{\mathrm{d}x}\Big|_{\text{极点}}=-\frac{1}{\varepsilon}\,\frac{\alpha_s}{2\pi}\,P_{qq}(x),\qquad P_{qq}(x)=C_F\Big[\frac{1+x^2}{1-x}\Big]_+.\tag{1.23}$$

它对 $x<1$ 逐点成立；连同 $\delta(1-x)$ 项，可由 Mellin 矩验证：$1/\varepsilon^2$ 相消，$1/\varepsilon$ 的系数为 $-\int_0^1\mathrm{d}x\,x^nC_F[(1+x^2)/(1-x)]_+$，对 $n=0,1,2,3$ 依次为 $0,\ \frac{4}{3},\ \frac{25}{12},\ \frac{157}{60}$（乘以 C_F，单位 $\alpha_s/2\pi$）。$n=0$ 即总截面，由夸克数守恒而有限；$n\ge1$ 的剩余极点由 $\overline{\mathrm{MS}}$ 碎裂函数吸收（第 9 章；参见 [218]）。一般地，对软辐射与共线分裂不敏感的观测量（红外安全量）逐阶有限 [76, 78, 79, 222]，最经典的例子就是下面的锥喷注截面。

1.1.8　Sterman–Weinberg 锥喷注截面

定义 1.6（两喷注事例 [76]）　若存在一对方向相反、半顶角为 δ 的锥，使锥外能量不超过 $\epsilon\sqrt{s}$，则称事例为两喷注事例。这里 ϵ 是总质心能量的分数，锥轴可任意选取。

由于 $-\vec p_k=\vec p_i+\vec p_j$ 位于 $\hat n_i$ 与 $\hat n_j$ 之间的大圆弧上，包含 $\hat n_i,\hat n_j,-\hat n_k$ 的最小锥半顶角为 $\theta_{ij}/2$。由此得到 图 1.4 中的判据（两个部分子都在锥外不可能，因为它们至少带走 $\sqrt{s}/2$）。三喷注区回避了全部奇异位形，其积分在 $D=4$ 收敛，所以 $\sigma_{2j}=\sigma_{\mathrm{tot}}-\sigma_{3j}$ 在维数正规化中是合法的。取 $y_{13}=$

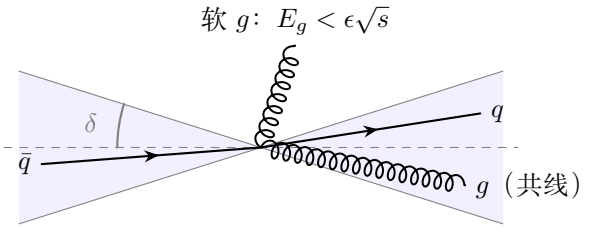


图 1.4　Sterman–Weinberg 两喷注区。在 $\mathcal{O}(\alpha_s)$，两喷注 $\iff$ 某对部分子夹角 $\theta_{ij} \le 2\delta$（三者都在锥内），或某部分子 $x_k < 2\epsilon$ 而另两者满足 $\pi - \theta_{ij} \le 2\delta$

$x_3 v$，$y_{23} = x_3(1-v)$（雅可比 x_3），利用精确恒等式 $M\,\mathrm{d}x_1\mathrm{d}x_2 = [N_0/(v(1-v)) - 2x_3^2]\,\mathrm{d}x_3\mathrm{d}v/x_3$，$N_0 = 1 + (1-x_3)^2$；共线条件为 $a < v < 1-a$，$a = d(1-x_3)/(1-x_3 d)$，$d = \sin^2\delta$。当 $\epsilon < \delta$ 时，$q\bar{q}$ 的非共线角不超过 $x_3(1+\mathcal{O}(x_3)) < 2\delta$，软条件恰好去掉 $x_3 < 2\epsilon$；软夸克区与 $q \parallel \bar{q}$ 区只给出 $\mathcal{O}(\epsilon \ln d)$ 与 $\mathcal{O}(d)$。于是

$$I_{3j} = \int_{2\epsilon}^{1} \frac{\mathrm{d}x_3}{x_3}\left[2N_0 \ln\frac{1-a}{a} - 2x_3^2(1-2a)\right] = 4\ln d\,\ln 2\epsilon + 3\ln d + \frac{2\pi^2}{3} - \frac{7}{2} + \mathcal{O}(\epsilon\ln d,\, d\ln\epsilon)\,, \tag{1.24}$$

其中用到 $\int_0^1 (2-2x+x^2)\frac{\ln(1-x)}{x}\mathrm{d}x = -\frac{\pi^2}{3} + \frac{5}{4}$。由 $\sigma_{3j} = \sigma_0 \frac{C_F\alpha_s}{2\pi} I_{3j}$ 及 式 (1.13)：

命题 1.7（Sterman–Weinberg 公式）　R 严格　当 $\epsilon < \delta \ll 1$（ϵ 为总能量的分数）时，

$$\sigma_{2j} = \sigma_0\Big\{1 - \frac{C_F\alpha_s}{\pi}\Big[4\ln\delta\,\ln 2\epsilon + 3\ln\delta + \frac{\pi^2}{3} - \frac{5}{2}\Big] + \mathcal{O}\big(\epsilon\ln\delta,\ \delta^2\ln\epsilon\big)\Big\}, \qquad C_F = \frac{4}{3}. \tag{1.25}$$

表 1.1　$c_{2j} \equiv (\sigma_{2j}/\sigma_0 - 1)/(C_F\alpha_s/\pi)$：精确数值积分（精确的区域边界、$v$ 积分取闭式、x_3 积分用 mpmath）与 式 (1.25) 的比较

(ϵ, δ)	数值	式 (1.25)（$\ln 2\epsilon$）	若误用 $\ln\epsilon$
$(10^{-4},\ 0.01)$	-143.8673	-143.8669	-156.635
$(10^{-5},\ 0.01)$	-186.2808	-186.2820	-199.050
$(10^{-6},\ 0.003)$	-288.2811	-288.2812	-304.388

注 1.8（关于常被引用的形式）　式 (1.25) 的括号与前因子 $\frac{4}{3}\frac{\alpha_s}{\pi}$ 与经典公式相同，但对数的宗量必须是 2ϵ（ϵ 为总能量的分数）。常见写法 “$4\ln\delta\ln\epsilon$” 只在 ϵ 表示束流能量 $\sqrt{s}/2$ 的分数时成立，此时 $\epsilon_{\text{束流}} = 2\epsilon$；否则相差 $\frac{4}{3}\frac{\alpha_s}{\pi}\cdot 4\ln 2\,|\ln\delta|$（表 1.1 最后一列）。公式要求 $\epsilon \lesssim \delta \ll 1$ 且锥轴可任意选取；当 $\epsilon \gg \delta$ 时，反冲（非共线）产生额外的 $\ln(\epsilon/\delta)$ 项，公式不再适用。$\mathcal{O}(\alpha_s^2)$ 的喷注截面见 [219, 220]。

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